

File Name: mux.vhd

Program 1: Design and Implementation of a 4-to-1 Multiplexer Using VHDL

```
C:/Users/USER/Desktop/Sayantan/mux.vhd - Default
Ln#
1      library IEEE;
2      use IEEE.STD_LOGIC_1164.ALL;
3
4      entity mux is
5      port(
6          X :in STD_LOGIC_VECTOR(3 downto 0);
7          S :in STD_LOGIC_VECTOR(1 downto 0);
8          Y :out STD_LOGIC
9      );
10     end mux;
11
12     architecture Behavioral of mux is
13     begin
14         process (X,S)
15         begin
16             case S is
17             when "00" => Y <= X(0);
18             when "01" => Y <= X(1);
19             when "10" => Y <= X(2);
20             when "11" => Y <= X(3);
21             when others => Y <= '0';
22             end case;
23         end process;
24     end Behavioral;
```

File Name: mux_8-1.vhd

Program 2: Design and Implementation of an 8-to-1 Multiplexer Using VHDL

```
C:/Users/USER/Desktop/Sayantan/mux_8-1.vhd - Default
Ln#
1
2      library IEEE;
3      use IEEE.STD_LOGIC_1164.ALL;
4
5      entity mux_8_1 is
6      port(
7          X :in STD_LOGIC_VECTOR(7 downto 0);
8          S :in STD_LOGIC_VECTOR(2 downto 0);
9          Y :out STD_LOGIC
10     );
11     end mux_8_1;
12
13     architecture Behavioral of mux_8_1 is
14     begin
15         process (X,S)
16         begin
17             case S is
18             when "000" => Y <= X(0);
19             when "001" => Y <= X(1);
20             when "010" => Y <= X(2);
21             when "011" => Y <= X(3);
22             when "100" => Y <= X(4);
23             when "101" => Y <= X(5);
24             when "110" => Y <= X(6);
25             when "111" => Y <= X(7);
26             when others => Y <= '0';
27             end case;
28         end process;
29     end Behavioral;
```

File Name: demux.vhd

Program 3: Design and Implementation of a 1-to-4 Demultiplexer Using VHDL

```
C:/Users/USER/Desktop/Sayantan/demux.vhd - Default
Ln#
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity demux is
5  port(
6      X :in STD_LOGIC;
7      S :in STD_LOGIC_VECTOR(1 downto 0);
8      Y :out STD_LOGIC_VECTOR(3 downto 0)
9  );
10 end demux;
11
12 architecture Behavioral of demux is
13 begin
14     process (X,S)
15     begin
16         Y<="0000";
17         case S is
18             when "00" => Y(0) <= X;
19             when "01" => Y(1) <= X;
20             when "10" => Y(2) <= X;
21             when "11" => Y(3) <= X;
22             when others => Y <= "XXXX";
23         end case;
24     end process;
25 end Behavioral;
26
```

File Name: demux_1-8.vhd

Program 4: Design and Implementation of a 1-to-8 Demultiplexer Using VHDL

```
C:/Users/USER/Desktop/Sayantan/demux_1-8.vhd (/demux_1_8) - Default
Ln#
2  library IEEE;
3  use IEEE.STD_LOGIC_1164.ALL;
4
5  entity demux_1_8 is
6  port(
7      X :in STD_LOGIC;
8      S :in STD_LOGIC_VECTOR(2 downto 0);
9      Y :out STD_LOGIC_VECTOR(7 downto 0)
10 );
11 end demux_1_8;
12
13 architecture Behavioral of demux_1_8 is
14 begin
15     process (X,S)
16     begin
17         Y<="00000000";
18         case S is
19             when "000" => Y(0) <= X;
20             when "001" => Y(1) <= X;
21             when "010" => Y(2) <= X;
22             when "011" => Y(3) <= X;
23             when "100" => Y(4) <= X;
24             when "101" => Y(5) <= X;
25             when "110" => Y(6) <= X;
26             when "111" => Y(7) <= X;
27             when others => Y <= "XXXXXXXX";
28         end case;
29     end process;
30 end Behavioral;
```

Figure 1: Simulated Waveform of a 4-to-1 Multiplexer

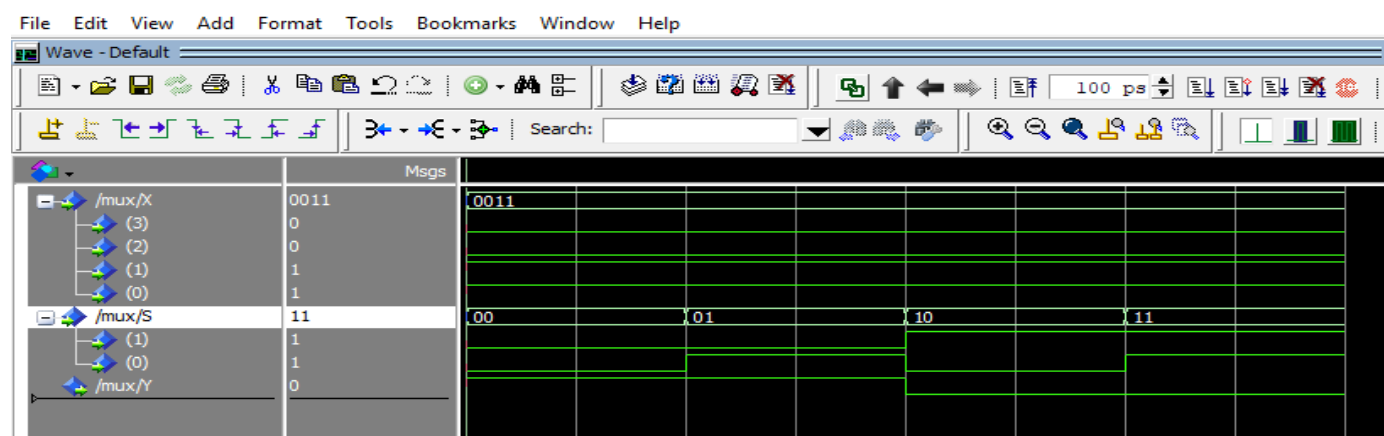


Figure 2: Simulated Waveform of an 8-to-1 Multiplexer

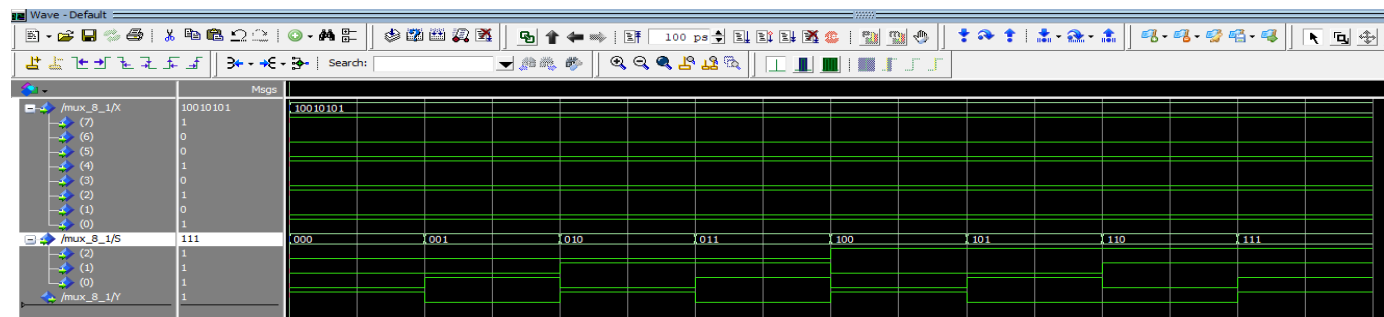


Figure 3: Simulated Waveform of a 1-to-4 Demultiplexer

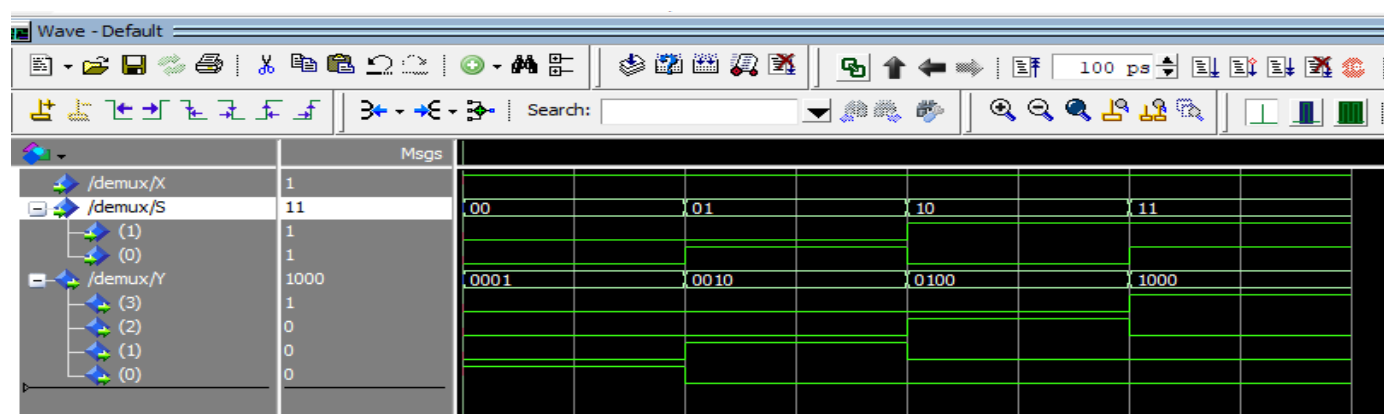


Figure 4: Simulated Waveform of a 1-to-8 Demultiplexer

